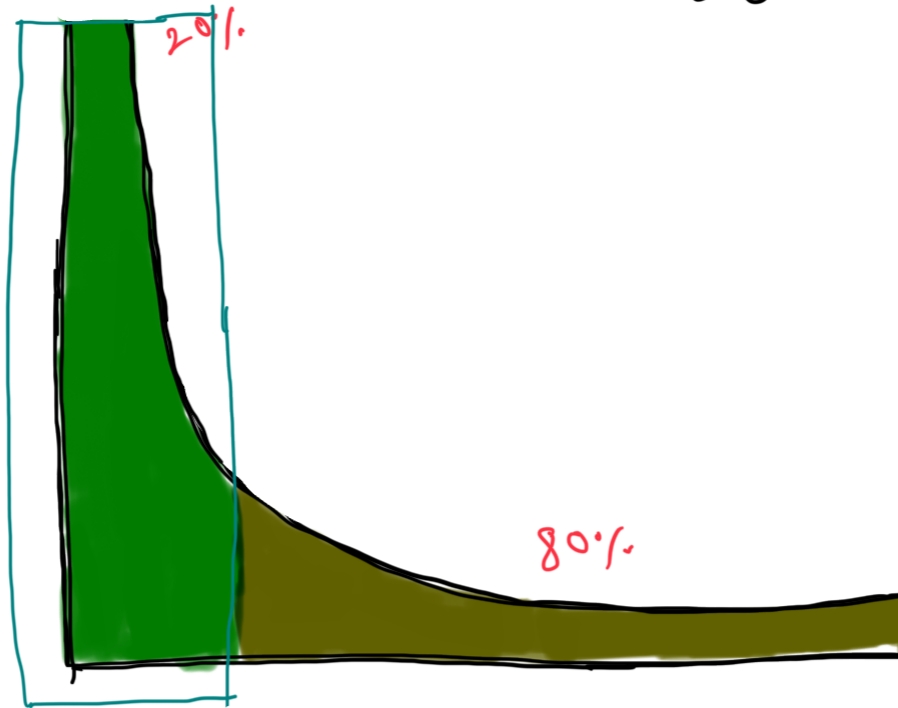


Power Law Distribution (Pareto Distribution)



This distribution is non-gaussian

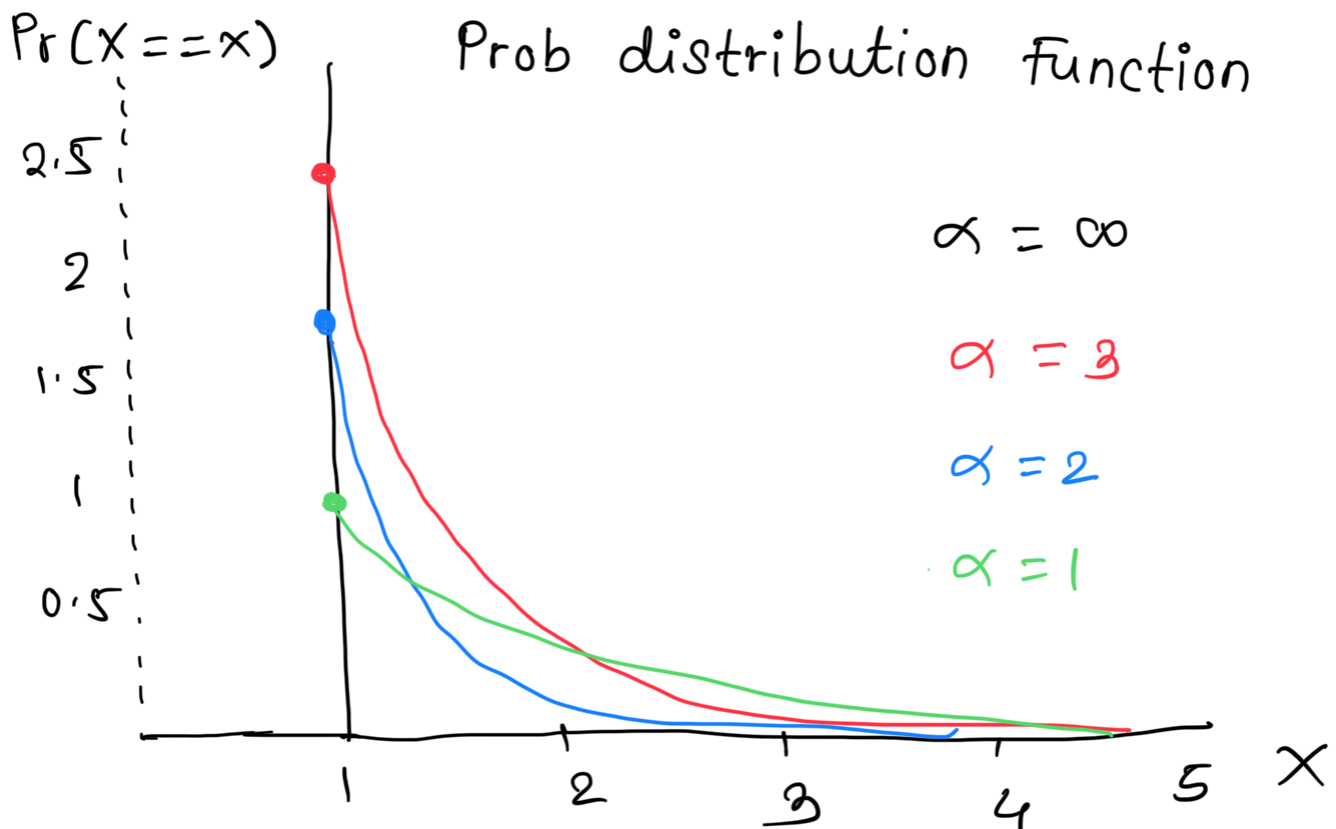
A relative change in one quantity resolve
in a proportional change in other
quantity

eg

80% sales are coming from 20% of
overall products.

80% of windows crashes are because of 20% of all the overall bugs

Pareto Distribution

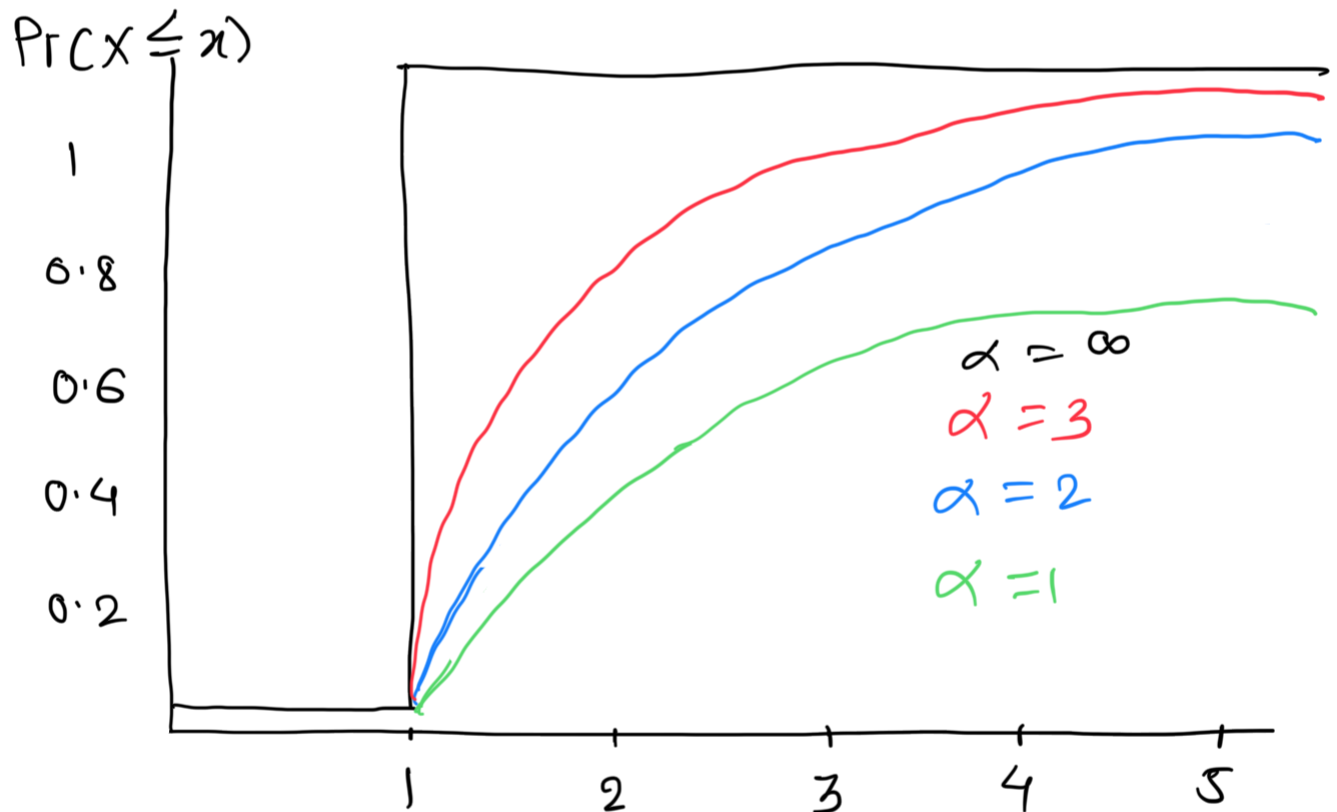


Pareto Type 1

Pareto Type 1 probability density Functions for various α with $x_m = 1$. As $\alpha \rightarrow \infty$, the distribution approaches $\delta(x - x_m)$,

where δ is the Dirac delta function

Cumulative Distribution Function



* Power Law Distribution means

- Small values happen very often
- Large values happen very rarely
- but they NEVER Fully disappear.

mathematically

$$P(X = x) \propto x^{-\alpha}$$

α = Positive constant causally between
2 and 3)

As x increases - prob drops very fast

* Key visual Features

- 1) Right skewed
- 2) Very long tail
- 3) Tail never ends
- 4) On log-log plot, it becomes a straight line

* Power law distribution means a system where a few extreme values dominate, there is no typical scale, and rare events matter more than averages.